

Can cash transfers improve the housing allocation?

Discussion note for Tommaso De Santo September 5, 2026

Answer. Yes, if the planner can commit to household-specific cash transfers when households are young and when they are old. In the stationary case described below, a small reform moves housing from old owners to young owners at every date. All household cohorts can retain their lifetime utility, and a resource surplus remains after accounting for every initial housing title. Prices and private choices clear every housing market along the path.

The permission matters. A promise to tax a household when old can support a transfer before its housing purchase. This provides public financing across ages, even though the private mortgage rule remains unchanged. The result establishes constrained inefficiency relative to this committed transfer class. It does not establish the same claim for a single round of cash gifts, or identify a welfare loss caused solely by households ignoring their effects on prices. No fertility externality is used.

1 The comparison

Use the existing household preferences and notation. A young household has adult consumption $x = c - \chi n$ and adult space $s = h - \kappa n$, with utility $\log x + \alpha \log s + \vartheta \log n$. Old utility is $\log c^2 + \gamma \log h^2 + \omega_B \log e$. Hold each household's fertility at its original choice and keep cohort sizes fixed. The exercise therefore compares housing use and the associated consumption and estates.

Start at a stationary equilibrium with mass $m > 0$ of owners and mass $m_R > 0$ of renters at each age. Within each tenure group, households have common income, liquid wealth, and fertility; their original ownership tastes can differ. Young owners have a strictly binding down-payment limit, positive saving, and a slack physical size cap. Old owners have slack retention and estate-composition bounds. Young and old renters are at their rental size cap. Tenure remains optimal in a sufficiently small neighborhood, as discussed below. This is a particular stationary case of the model, not a theorem for every entrant distribution or every set of binding constraints.

Write $u = qr = (1 + q\tau^p - q)P$ for the stationary cost of housing services. Use h^Y, h^O, c^O for stationary young-owner housing, old-owner housing, and old-owner consumption. Define three preference sums and the young owner's housing value in units of consumption:

$$A = 1 + \gamma + \omega_B, \quad \rho = 1 + \beta A, \quad B = 1 + \beta(1 + \omega_B), \quad MV^Y = \frac{\alpha x}{s}, \quad s = h^Y - \kappa n. \quad (1)$$

The original saving and old-housing conditions give

$$c^O = \frac{\beta x}{q}, \quad h^O = \frac{\gamma \beta x}{qu}, \quad MV^O = u. \quad (2)$$

Thus $MV^Y > u$ is the housing-value gap generated by the binding purchase limit.

Transfers and ownership. A young household receives cash ℓ_t^Y before closing, and a previously announced payment ℓ_{t+1}^O when old. Negative payments are enforceable lump-sum taxes. A household receives its announced payments under either tenure choice; they are not housing subsidies. The young-owner limit remains $(1 - \phi)P_t h_t^Y \leq b + \ell_t^Y$. Private saving remains nonnegative. The ordinary property-tax rebate T_t retains its original timing and budget.

We must also identify the owners of the initial titles not held by current old households. For this proposition, assign the remaining $R = \bar{H} - mh^Y$ titles to a passive outside owner. This owner has no domestic housing demand, sells its initial titles at the reform price, and can trade the world bond subject to its present-value budget. It values additional goods and receives the residual of the cash-transfer account at each date. Its title revaluation and its entire transfer stream enter the welfare comparison. This is an explicit completion of the original model's unspecified initial ownership, not a claim that zero intermediary profits eliminate initial owners. The planner can enforce the announced payments to and from this owner. With sufficiently small reforms and slack outside credit capacity, all promised payments are feasible without a Ponzi scheme.

2 A fully clearing improving path

Proposition 1 (Committed cash transfers). *Under the preceding assumptions, suppose $MV^Y > u$ and*

$$\lambda = \frac{\alpha h^O}{\rho s} < 1. \quad (3)$$

Then a sufficiently small committed cash-transfer reform is a Pareto improvement. Young owners occupy more housing and old owners less at every finite date. All household cohorts can be indifferent and the passive initial title owner strictly better off. Every housing market and each date's transfer budget balance. The allocation converges to the original steady state.

The convergence condition limits how strongly a small consumption change in one cohort changes the next cohort's compensated housing choice. It is a sufficient condition for this construction, not a claim about every policy transition in the model.

First date. Let $\epsilon > 0$ be a small proportional increase in the current service cost. Set

$$P_0 = P + \frac{u\epsilon}{1 + q\tau^p}, \quad P_t = P \quad (t \geq 1). \quad (4)$$

This implies $u_0 = u(1 + \epsilon)$ and $u_t = u$ thereafter. The price path is supported by the transfers below; it is not a price control. Let $\eta = (1 + \omega_B)/A$. Exact compensation of the initial old owners gives

$$c_0^O = c^O(1 + \epsilon)^{\gamma/A}, \quad h_0^O = h^O(1 + \epsilon)^{-\eta}, \quad e_0^O = \omega_B c_0^O / q. \quad (5)$$

These are their optimal private choices at the new user cost. Their utility is unchanged, and they release housing. Keep renter consumption and housing unchanged. Market clearing then requires

$$h_0^Y = h^Y + h^O - h_0^O, \quad x_0 = x \left(\frac{s}{h_0^Y - \kappa n} \right)^{\alpha/\rho}. \quad (6)$$

The young owner has more space and less adult consumption, with the same lifetime utility. Its old-age consumption and estate adjust through saving.

Every later date. Define $X_t = x_t/x$. The preceding young cohort chooses old housing $h^O X_{t-1}$ at the unchanged future user cost. Clearing housing and preserving the next young cohort's lifetime utility give

$$h_t^Y = h^Y + h^O(1 - X_{t-1}), \quad X_t = \left[1 + \frac{h^O}{s}(1 - X_{t-1}) \right]^{-\alpha/\rho}, \quad t \geq 1. \quad (7)$$

Since $X_0 < 1$ and the derivative of this map is at most $\lambda < 1$ on $X \leq 1$, the sequence increases to one. Young housing is above its original level at every date, old housing is below it, and both converge to baseline. All sufficiently small changes preserve the strict private constraints stated above. This recursion verifies the infinite continuation rather than fixing future prices without clearing future markets.

Private implementation. Pay each young owner the closing cash needed for its chosen home. Choose its committed old-age payment from its original lifetime budget:

$$\ell_t^Y = (1 - \phi)P_t h_t^Y - b, \quad (8)$$

$$\ell_t^Y + q\ell_{t+1}^O = \rho x_t + u_t h_t^Y + \chi n - y - b - T_t - qT_{t+1}. \quad (9)$$

The saving Euler equation, old demand conditions, and binding purchase limit then support the proposed allocation. A small change leaves the young housing multiplier positive. Cash payments to initial old households follow from their original dated budgets. Initial old renters and young renters receive $(u_0 - u)h_R^{\max} - (T_0 - T)$; later renters need no transfer. Each date, the passive owner receives the negative of the other groups' aggregate net transfers, so the government issues no debt. The passive owner can have payments of either sign across dates. Its access to the world bond, together with enforceable future household taxes, is the financial capacity that supports this construction. Date-by-date government balance alone would not prove that this financing is feasible.

3 The resource gain and its interpretation

Let ΔW be the change in the passive owner's present-value resources, including $R(P_0 - P)$ and its complete transfer stream. Summing the original dated budgets, using housing clearing and the property-tax rebate rule, gives the exact identity

$$\Delta W = -m \left[(1 + \omega_B)(c_0^O - c^O) + B \sum_{t=0}^{\infty} q^t (x_t - x) \right]. \quad (10)$$

Every initial title is counted. House-price revaluations and domestic taxes cancel after consolidation; changes in consumption and warm-glow estates remain. The negative consumption changes of successive young cohorts can pay for initial old-owner compensation.

Differentiate the constructed path at $\epsilon = 0$. Equations (5)–(7) imply $dh_t^Y/d\epsilon = h^O \eta \lambda^t$ and $dx_t/d\epsilon = -(MV^Y/\rho)h^O \eta \lambda^t$. Substituting these in (10) and using (2) yields

$$\left. \frac{d\Delta W}{d\epsilon} \right|_0 = \frac{mh^O \eta}{1 - q\lambda} (MV^Y - u) > 0. \quad (11)$$

The gain has the desired source: the housing is worth more to the young than to the old. All household utilities have already been compensated, so the remaining positive resources establish a strict Pareto improvement. The proof assigns the surplus to the passive owner for convenience; no welfare weights or fertility benefits enter the argument.

The initial young owners can also gain strictly. At a fixed small reform, raise x_0 slightly above (6), keeping $x_0 < x$, and use the same recursion from date one onward. Their housing is unchanged and their lifetime utility rises. By continuity, some positive resource surplus remains, all later cohorts retain their utility, and housing still moves toward the young.

Real transfer costs. The proposition uses zero additional real cost. An explicit extension charges $C(h_t^Y - h^Y)$ goods per affected owner at date t , with $C(0) = 0$ and $C'(0) = K$. Deduct these costs from the passive owner’s cash account. The derivative in (11) then replaces $MV^Y - u$ with $MV^Y - u - K$. This is a cost of the extra housing reassigned at each date, not automatically a fixed moving cost or a cost of the change from the previous date.

Tenure choice. The transfers are fixed by household identity before the new choice. Holding fertility fixed means an original owner considering renting must use its own n , and conversely. If the two tenures originally choose different fertility levels, a household at the original taste cutoff can have a strict preference for its original tenure in this comparison. A uniform positive margin and continuity preserve tenure for a sufficiently small reform, including a continuous taste distribution. This condition must be checked rather than inferred from a fixed owner share.

4 Why a one-time gift is a different question

A transfer only at the initial date cannot generally supply the compensation in (9) for all later cohorts. Their housing demand changes as the initial cohort ages, so prices adjust again. In the illustrative stationary case, the linearized equilibrium after transfers cease has one converging price root, -0.05518 , and one divergent root, 3.30801 . The converging price response alternates in sign. With renters at their housing caps, their lifetime welfare also alternates in sign, so a nonzero small converging price response hurts some future cohorts.

The separate review derives the nonlinear local argument and treats the zero-price-response case using the complete resource account. Its conclusion is a local obstruction in this stationary regime. It is not a global impossibility theorem for one-time transfers under all preferences, initial conditions, or tenure responses. The derivative roots alone would not establish that stronger claim. The earlier fixed-future-price, four-household examples omit both this continuation and some initial title owners; they cannot settle the complete-economy comparison.

Recommendation. Use the committed-transfer result if we accept public financing across ages as part of the constrained planner’s powers. State that permission before the proposition. If the intended planner is restricted to one initial cash redistribution, the present model does not yet supply the desired general result, and the local obstruction is substantive. No new fertility motive is needed to obtain the positive theorem above.

5 Verification

Independent derivations, original-budget accounting, and twelve direct household optimizations support the proof. Accounting errors are below 9×10^{-16} ; optimized choices match within 1.8×10^{-7} . The minimum tenure margin exceeds 0.0357 and $\lambda = 0.26088$. These are illustrative checks, not calibration evidence. The reviews and receipts are indexed in `output/model/simplified_olg_amendments/README.md`. Reproduce the checks, including independent optimizations with SciPy, with

```
python3 code/model/tools/verify_simplified_olg_planner_benchmarks.py --original-optimizers
```

The author-controlled manuscript and quantitative model were not edited.